

Jay Padhara

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Summary

Software Engineer evolving into a Data Analyst and Machine Learning professional, currently pursuing a Master's in Machine Learning and Data Analytics at Aalen University. Strong background in software development with hands-on experience in mobile and backend systems using React Native, Node.js, and APIs. Skilled in Python, data preprocessing, exploratory data analysis (EDA), and building basic predictive models. Focused on transforming raw data into actionable insights and developing data-driven solutions that support intelligent decision-making in applications.

Education

Hochschule Aalen

MSc Machine Learning and Data Analytics

Aalen, Baden-Württemberg

March 2026 – Present

- **Coursework:** Machine Learning, Deep Learning, Artificial Intelligence, Natural Language Processing (NLP), Data Analytics, Statistical Modelling, Data Mining, Data Engineering, Python, Model Evaluation, Feature Engineering.

The Maharaja Sayajirao University of Baroda

Bachelor of Computer Applications

Vadodara, India

March 2021 – April 2024

- **Coursework:** Python, C, C++, Data Structures and Algorithms, Database Management Systems (SQL), Computer Networks, Operating Systems, Web Development (HTML, CSS, JavaScript), Object-Oriented Programming (OOP).

Professional Experience

Vasundhara Infotech

React Native Developer

Surat, India

April 2024 – February 2026

- Owned full product delivery with **Expo EAS & CI/CD automation**, accelerating reliable production releases..
- **Optimized scalable mobile architecture**, cutting app load time **25% reduction in application load times** production applications.
- Built high-performance Python & REST integrations for concurrency, resilience, and maintainable systems.

Implies Solutions

React Native Developer – Intern

Surat, India

December 2023 – March 2024

- **Researched AI/ML integrations**, connecting mobile experiences with intelligent backend systems.
- **Translated complex product requirements** into scalable API solutions within Agile delivery cycles.
- **Delivered 2+ production releases**, strengthening app stability through testing and rapid issue resolution.

Projects

Insurance Cost Prediction

- Engineered an end-to-end machine learning pipeline on **1,300+ data samples**, including data profiling, cleaning, feature engineering, transformation, and normalization for structured datasets.
- Benchmarked and optimized regression models (Linear Regression, Random Forest) improving predictive performance by **~12–18%** through error analysis and performance tuning techniques.
- Derived actionable insights through statistical analysis and visual storytelling, strengthening understanding of model behaviour and generalization.

Heart Disease Prediction

- Designed a predictive classification model using patient health indicators, processing **1,000+ records** with data conditioning, feature encoding, normalization, and scaling techniques to prepare structured medical datasets for modelling.
- Evaluated and refined models (Logistic Regression, Decision Tree, Random Forest) through cross-validation and performance diagnostics.
- Interpreted model outputs using feature importance and statistical insights to understand key risk factors influencing predictions.

Champions Spirit

- Integrated REST APIs and backend services supporting **real-time scheduling between 200+ users and 30+ trainers**, enabling seamless session booking, trainer availability tracking, and user data synchronization.

- Improved UI/UX by implementing intuitive navigation, responsive components, and user-centric features that simplified fitness exploration and session booking.
- Optimized app performance and scalability, focusing on smooth transitions, efficient API handling, and reusable component architecture to support diverse user activities.

YourPetDiary

- Conceived a mobile platform for pet enthusiasts to log wellness metrics, vaccination timelines, and real-time pet movements, centralizing key pet insights.
- Built an integrated marketplace handling **1,000+ product entries** and enabling seamless browsing and purchasing of pet products, complete with order tracking and inventory management.
- Crafted features for sharing pet profiles, recording daily routines, and connecting with fellow owners, promoting interaction and knowledge exchange.

Knowledge & Skills

Programming Languages: Python, JavaScript, TypeScript

Data & Machine Learning: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, TensorFlow, PyTorch, BeautifulSoup

Data Concepts: EDA, Data Cleaning, Feature Engineering, Model Evaluation, Statistical Analysis, Data Visualization

Backend & APIs: Node.js, Express.js, REST APIs, Firebase, MongoDB, SQL, TanStack Query, WebSocket, Zustand, WatermelonDB

DevOps & Tools: Git, GitHub, CI/CD (Expo EAS Build)

Mobile & Frontend: React Native, React.js, Expo, Redux, HTML, CSS, Android Studio, XCode (iOS), Native Modules, Context API, Testing Library

Volunteer Engagement

BAPS (Bochasanwasi Akshar Purushottam Swaminarayan Sanstha)

Volunteer

2015 – Present

- Actively contributed to the planning and execution of cultural events with **1000+ devotees**.
- Collaborated with cross-functional volunteer teams to ensure smooth event operations and logistics.
- Supported coordination of on-ground activities, improving organizational efficiency and execution flow.

Placement Cell — The Maharaja Sayajirao University of Baroda

Placement Cell Coordinator

2021 – 2024

- Acted as a key liaison between university students and external recruiters, including HR teams and hiring managers.
- Coordinated end-to-end campus recruitment processes, including scheduling interviews, drives, and communication flow.
- Managed professional communication with multiple companies to ensure smooth execution of placement activities.

Languages

English: C1 (Business Fluent)

German: A2 (Improving)